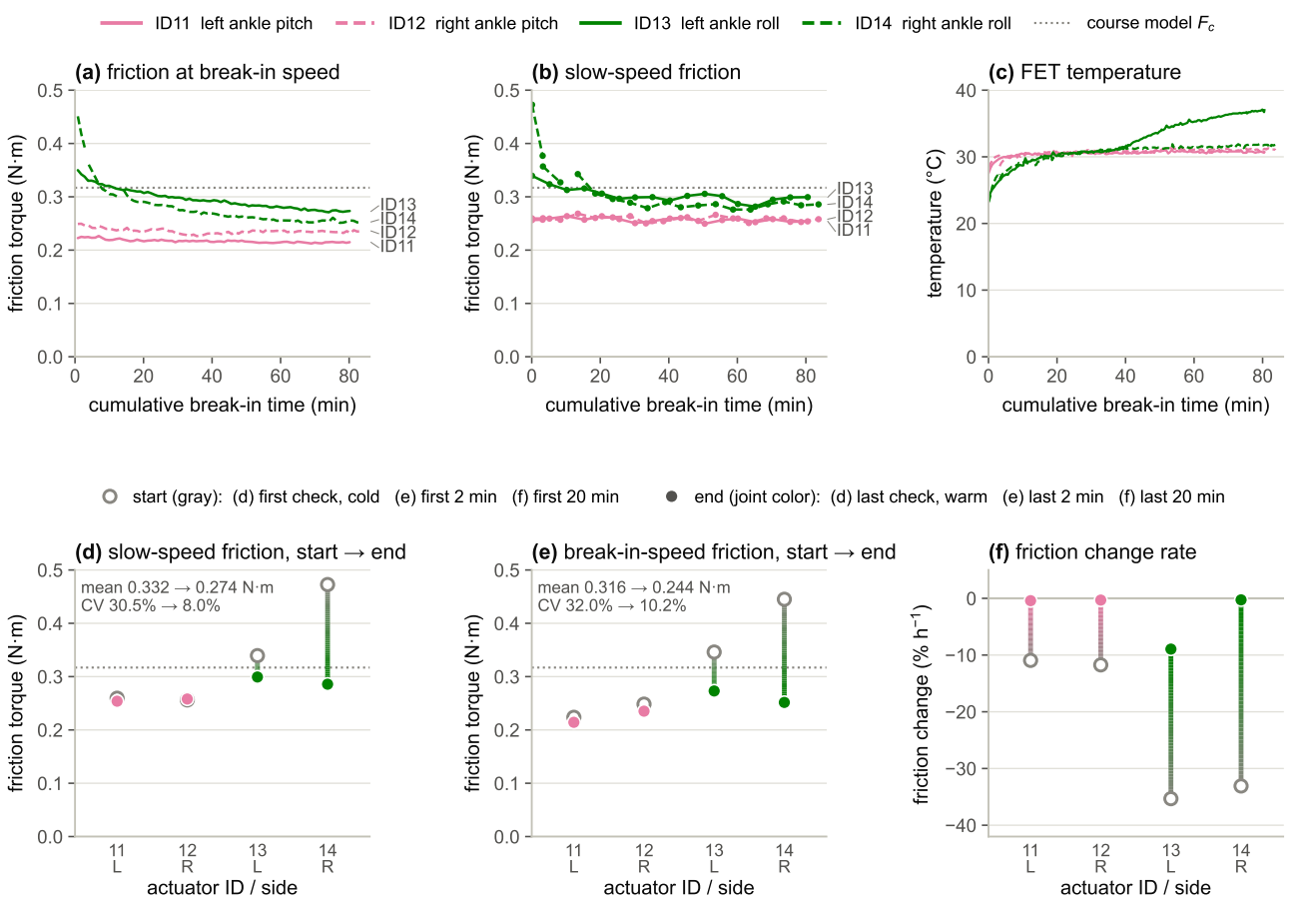

ID11 left ankle pitch
ID12 right ankle pitch
ID13 left ankle roll
ID14 right ankle roll
course model F_c
(a) friction at break-in speed
friction torque (N·m)
cumulative break-in time (min)
ID13
ID14
ID12
ID11
(b) slow-speed friction
friction torque (N·m)
cumulative break-in time (min)
ID13
ID14
ID12
ID11
(c) FET temperature
temperature (°C)
cumulative break-in time (min)
start (gray): (d) first check, cold (e) first 2 min (f) first 20 min
end (joint color): (d) last check, warm (e) last 2 min (f) last 20 min
(d) slow-speed friction, start → end
mean 0.332 → 0.274 N·m
CV 30.5% → 8.0%
friction torque (N·m)
11 L
12 R
13 L
14 R
actuator ID / side
(e) break-in-speed friction, start → end
mean 0.316 → 0.244 N·m
CV 32.0% → 10.2%
friction torque (N·m)
11 L
12 R
13 L
14 R
actuator ID / side
(f) friction change rate
friction change (% h⁻¹)
11 L
12 R
13 L
14 R
actuator ID / side